

VCS Module

VMD0054

ESTIMATING LEAKAGE FROM THE DISPLACEMENT OF AGRICULTURAL ACTIVITIES

Version 1.1

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Sectoral Scope 14: Forestry and Other Land Use

Sectoral Scope 15: Agriculture

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CONTENTS

1	SUMMARY DESCRIPTION	4
2	SOURCES	4
3	DEFINITIONS	5
4	APPLICABILITY CONDITIONS	6
5	PROCEDURES	6
5.1	Step 1: Determine Change in Production in Project Area	6
5.2	Step 2: Determine Impact of Leakage Mitigation Activities (Optional)	8
5.3	Step 3: Determine Land Impact	12
5.4	Step 4: Determine Change in Carbon Stocks in New Lands Brought into Production	14
5.5	Step 5: Determine Leakage Emissions	16
6	DATA AND PARAMETERS	17
6.1	Data and Parameters Available at Validation	17
6.2	Data and Parameters Monitored	20
7	REFERENCES	29
	APPENDIX 1: BACKGROUND ON DEFAULT VALUES	30
	DOCUMENT HISTORY	32

1 SUMMARY DESCRIPTION

Projects applying VCS methodology *VM0047 Afforestation, Reforestation, and Revegetation* must use this module to estimate leakage. Projects applying *VM0042 Improved Agricultural Land Management* or *VM0032 Methodology for the Adoption of Sustainable Grasslands through Adjustment of Fire and Grazing* may use this module where permitted by the respective methodology.

This module accounts for leakage related to the displacement of pre-project agricultural activities (including grazing) and fuelwood collection activities. It includes activity-shifting leakage caused by the baseline agent and market leakage caused by other actors.

The module also accounts for leakage mitigation. Leakage mitigation reduces the amount of prior production in the project area that is subject to leakage. Leakage mitigation may occur outside the project area where the project proponent supports interventions to increase productivity through agricultural intensification, or through cross-commodity leakage mitigation where new commodities are produced in the project area.

The module estimates net leakage based on three factors:

- **Leakage:** Displacement of agricultural activities inside the project area to locations outside of the project area as a result of project activities. This module calculates the proportion of displaced production that results in clearing of previously uncultivated lands and subsequent emission of greenhouse gases (GHGs).
- **Cross-commodity production:** Introduction of new agricultural commodities to the project area, which can reduce the amount of displaced production. Any incremental increase in GHG emissions resulting from cross-commodity production is accounted for as project emissions.
- **Leakage mitigation:** Increased production of pre-project commodities outside of the project area through intensification. This reduces leakage but requires accounting of the incremental increase in GHG emissions from leakage mitigation activities.

2 SOURCES

This methodology uses the most recent version of VM0047 Afforestation, Reforestation, and Revegetation.

The following Clean Development Mechanism (CDM) tool has informed the development of this module:

- *AR-TOOL15 Estimation of the Increase in GHG Emissions Attributable to Displacement of Pre-project Agricultural Activities in A/R CDM Project Activity, v2.0*

3 DEFINITIONS

In addition to the definitions set out in the most recent version of the *VCS Program Definitions*, the following definitions apply to this module.

Agricultural activity

Activity resulting in the production of an agricultural commodity on agricultural land

Agricultural commodity

Any (part of) plant, animal, or animal product, produced on agricultural land

Cross-commodity production

Production of an agricultural commodity that is introduced into the project area as part of the project activity. Differs from pre-project commodity production and is accounted for as a change in monitored production.

Fuelwood

Wood used as fuel, including firewood or charcoal produced from woody biomass

Land sparing

Net decrease in land area required for agricultural production due to an increase in the production of a commodity introduced into the project area; assumed to have occurred when the value for net new land brought into production is negative

Leakage mitigation

Observed increase in commodity production above a historic baseline, that results from clearly identifiable leakage mitigation activities occurring outside of the project area (e.g., agricultural intensification)

Leakage mitigation activity start date

Date on which an activity that increases commodity production for the purpose of leakage mitigation is implemented outside of the project area

Subsistence commodity

Agricultural commodity that is not destined for commercial markets

4 APPLICABILITY CONDITIONS

Project proponents must meet all applicability conditions of the version of VM0032, VM0042, or VM0047 that is applied to the project.

Leakage may only be deemed de minimis where the project proponent demonstrates that lands enrolled in the project have not been used for agricultural activities or fuelwood production in the 10 years immediately preceding the project start date. This must be evidenced by remote sensing analysis of land use and attestations from landowners or managers.

5 PROCEDURES

Leakage emissions from project activities are calculated for the entire project area and for at least five years from the project start date.¹ For grouped projects, leakage must be calculated for all instances for five years after the most recent addition of an instance.

5.1 Step 1: Determine Change in Production in Project Area

Baseline agricultural activities and fuelwood commodity production in the project area must be estimated using historical production records for each commodity during the historical reference period. Use the process in Sections 5.1.1–5.1.3 to estimate historical production for each commodity.

5.1.1 Step 1.1: Determine the Historical Reference Period

The historical reference period must be the longer of the following:

- 1) The three-year period immediately preceding the project or instance start date
- 2) One complete crop rotation

¹ Displacement of pre-project agricultural activities and fuelwood collection is treated as a one-time occurrence, and leakage that has already been accounted for during a verification event is not counted again. Leakage from displacement is calculated over a period of at least five years to capture the incremental changes in foregone production resulting from assumed increases in yield. The incremental increase does not result in repeated or annual land conversion. The leakage calculation provides a conservative estimate of leakage where the baseline level of production is expected to increase over time.

5.1.2 Step 1.2: Estimate Baseline and Monitored Production of Each Commodity

Identify all commodities for which production will be impacted as part of project activities (displaced pre-project production or introduced cross-commodity production).

Use historical and project data, as appropriate, to quantify the change in production of each commodity. Eligible types of historical production records are listed below. Data sources must be used in the following order, where available and verifiable (e.g., where multiple data sources are available, the data source nearest to the top of the list must be used):

- 1) Grower records (e.g., management logs, receipts or invoices, logs or files containing machine and/or sensor data)
- 2) Remote sensing approaches where methods have been calibrated and validated in a similar region and for the commodities produced
- 3) Most recent national or sub-national average values derived from published census data for similar ownership type and land holdings (e.g., smallholders on private lands)
- 4) For fuelwood, IPCC regional aboveground biomass growth rates (where regional or national averages are unavailable)

In Equation (1), baseline production for commodity j in year t is calculated as the average annual historical production, adjusted for a fixed annual increase in production.

$$BP_{j,t} = \frac{\sum_{h=1}^H p_{j,h}}{H} \times (1 + r_j)^t \quad (1)$$

Where:

$BP_{j,t}$	=	Baseline production in the project area of commodity j in year t (units of production)
$p_{j,h}$	=	Production in the project area of commodity j in year h of the historical reference period (units of production)
H	=	Duration of historical reference period (years)
r_j	=	Annual growth rate of yield for commodity j (%)
t	=	1, 2, 3, ..., t years elapsed since the project start date

Note – For new cross-commodity production introduced as part of the project activity, baseline production ($BP_{j,t}$) is set to zero as there would be no prior production of the introduced commodity.

Monitored production of commodity j in year t is determined as the average annual production observed in the project area in year t .

$$MP_{j,t} = \frac{\sum_{t=1}^t p_{j,t}}{W} \quad (2)$$

Where:

$MP_{j,t}$	= Monitored production in the project area of commodity j in year t (units of production)
$p_{j,t}$	= Production in the project area for commodity j in year t of the project period (units of production)
W	= Duration of current monitoring period (years)

5.1.3 Step 1.3: Estimate Change in Commodity Production

The change in commodity production must be calculated for commodities produced in the project area during the historical reference period. Where cross-commodity production is applied, the change in production may be calculated for new commodities introduced as part of the project activity. The change in production is the difference between baseline and monitored production for each commodity.

$$CP_{j,t} = BP_{j,t} - MP_{j,t} \quad (3)$$

Where:

$CP_{j,t}$	= Change in production in the project area of commodity j in year t (units of production)
$BP_{j,t}$	= Baseline production in the project area for commodity j in year t (units of production)
$MP_{j,t}$	= Monitored production in the project area of commodity j in year t (units of production)
t	= 1, 2, 3, ..., t years elapsed since the project start date

5.2 Step 2: Determine Impact of Leakage Mitigation Activities (Optional)

Project proponents may implement activities outside the project area that reduce the amount of leakage. The areas in which such activities take place are called leakage mitigation areas. Where leakage mitigation activities are implemented, the project proponent must:

- 1) ensure the leakage mitigation area is located outside of the project area.
- 2) ensure that the leakage mitigation area is in the same country as the project.

- 3) secure exclusive claim to a specific amount of increased commodity production within a leakage mitigation area through an agreement with the landowner or manager responsible for implementing leakage mitigation activities. The agreement must specify the amount of increased production attributable to the leakage mitigation activities, as a percentage or absolute value.
- 4) account for leakage mitigation only in the years during which an agreement is in place. The terms of the agreement must be no less than five years.

Note – Leakage mitigation may occur on land owned or managed by individuals or groups participating in the project, but land used for leakage mitigation must not be included in the project area.

- 5) account for significant increases in GHG emissions associated with leakage mitigation activities. All projects applying leakage mitigation activities must use Appendix 2 of VM0047 to identify significant emission sources and report any net increase attributable to leakage mitigation activities (e.g., fertilizers, fossil fuel consumption). Where supplemental feeding is used, account for emissions from increased production of supplemental feedstocks.
- 6) not exceed maximum carrying capacity nor deplete vegetation and soil resources where leakage mitigation involves livestock intensification. Project proponents must demonstrate the maximum carrying capacity with reports prepared by, or signed attestations from, qualified independent local experts (e.g., agricultural extension agent, professional agronomist).
- 7) where such activities include fuelwood production, ensure that this only involves the establishment of new tree plantations.

The impact of leakage mitigation activities is determined using the steps outlined in Sections 5.2.1–5.2.2.

5.2.1 Step 2.1: Estimate Change in Production in Leakage Mitigation Areas

Leakage mitigation ($LM_{j,t}$) is quantified as the increase above baseline in production of commodity j in year t in the leakage mitigation area. Baseline production in the leakage mitigation area ($OBP_{j,t}$) is based on production before the leakage mitigation activity began. The leakage mitigation historical reference period is the longer of the following:

- The three years immediately prior to the leakage mitigation activity start date
- One complete crop rotation (where applicable)

Baseline production of agricultural commodities in the leakage mitigation area is the average annual historical production, adjusted with an annual growth rate and calculated as follows:

$$OBP_{j,t} = \frac{\sum_{k=1}^K op_{j,k}}{K} \times (1 + r_j)^t \quad (4)$$

Where:

- $OBP_{j,t}$ = Baseline production in the leakage mitigation area for commodity j in year t (units of production)
- $op_{j,k}$ = Baseline production in the leakage mitigation area for commodity j in year k of the leakage mitigation historical reference period (units of production)
- K = Duration of leakage mitigation historical reference period (years)
- r_j = Annual growth rate of yield for commodity j (%)
- t = 1, 2, 3, ..., t years elapsed since the project start date

Baseline production of fuelwood in the leakage mitigation area ($OBP_{fuelwood,t}$) is zero, as only newly established tree plantations are eligible.

Leakage mitigation is equal to the monitored production in the leakage mitigation area minus the baseline levels of production in that area, and is calculated as follows:

$$LM_{j,t} = OMP_{j,t} - OBP_{j,t} \quad (5)$$

Where:

- $LM_{j,t}$ = Leakage mitigation for commodity j in year t (units of production)
- $OMP_{j,t}$ = Monitored production in the leakage mitigation area for commodity j in year t (units of production)
- $OBP_{j,t}$ = Baseline production in the leakage mitigation area for commodity j in year t (units of production)
- t = 1, 2, 3, ..., t years elapsed since the project start date

The net change in production that may result in leakage is calculated as the change in production minus any leakage mitigation:

$$l_{j,t} = CP_{j,t} - LM_{j,t} \quad (6)$$

Where:

- $l_{j,t}$ = Net change in production subject to leakage for commodity j in year t (units of production)

$CP_{j,t}$	= Change in production in the project area of commodity j in year t (units of production)
$LM_{j,t}$	= Leakage mitigation for commodity j in year t (units of production)
t	= 1, 2, 3, ..., t years elapsed since the project start date

5.2.2 Step 2.2: Quantify Emissions from Leakage Mitigation Activities

Leakage mitigation activities involving agricultural intensification in the leakage mitigation area can potentially increase GHG emissions (e.g., through increased fertilizer use). Project proponents must account for all significant GHG emissions from leakage mitigation activities. Regardless of the methodology applied to the project, project proponents must use Appendix 2 of VM0047 to determine whether decreases in carbon pools and increases in GHG emissions are de minimis.

$E_{LM,t}$ is the cumulative leakage emissions through time t , calculated by summing emissions from fertilizer, fossil fuels, and other sources from the start of mitigation activities ($t = 0$) to the present (t):

$$E_{LM,t} = \sum_{t=0}^t (E_{fert,t} + E_{FF,t}) \quad (7)$$

Where:

$E_{LM,t}$	= Total leakage mitigation emissions in year t (t CO ₂ e)
$E_{fert,t}$	= Leakage mitigation emissions from incremental increase in fertilizer use in year t (t CO ₂ e)
$E_{FF,t}$	= Leakage mitigation emissions from incremental increase in fossil fuel use in year t (t CO ₂ e)
t	= 1, 2, 3, ..., t years elapsed since the project start date

Where fossil fuel use increases due to leakage mitigation activities, emissions from fossil fuel consumption are summed over all fuel types i and commodities j , as follows:

$$E_{FF,t} = \sum_{i,j} (F_{i,j} \times EF_i) \quad (8)$$

Where:

$E_{FF,t}$	= Leakage mitigation emissions from incremental increase in fossil fuel use in year t (t CO ₂ e)
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$F_{i,j}$	=	Quantity of fossil fuel i consumed due to leakage mitigation activity for commodity j (volume unit)
EF_i	=	Emission factor for fossil fuel i (dimensionless)
t	=	1, 2, 3, ..., t years elapsed since the project start date

For all projects, leakage mitigation emissions from incremental increases in fertilizer use ($E_{fert,t}$) must be calculated using equations found in VM0047, *Section 8.2.2.2 Area-based Quantification of Emissions from Fertilizer Application* and applying relevant data from the leakage mitigation area.

Assessment of fertilizer emissions or demonstration of stocking rates compared to carrying capacity is not required where the leakage mitigation area is included in a registered carbon project and changes in GHG emissions and soil carbon stocks are assessed.

5.3 Step 3: Determine Land Impact

This step estimates the area of land outside of the project area that is brought into production due to displaced production (positive values) or spared due to cross-commodity production (negative values). The net change in production for each commodity is converted to an area-based estimate. This calculation uses assumed default values for increased supply (IS_j) and new lands (NL_j) and regionally specific land yield data:

- **Increased supply (IS_j):** proportion of displaced production that is replaced by increased supply
- **New lands (NL_j):** proportion of IS produced on new lands
- **Expected yield ($G_{j,t}$):** units of commodity j per hectare in year t , established using regionally appropriate data

The default assumptions in Table 1 are made based on commodity type (agricultural or fuelwood) and selection of leakage-only or cross-commodity production accounting.

Table 1. Default values for increased supply (IS_j) and new lands (NL_j)

Default values	Leakage-only default (%)		Cross-commodity production default (%)	
	Displaced agricultural commodity	Displaced fuelwood commodity	Displaced agricultural commodity	Introduced agricultural commodity
Increased supply (IS_j)	75	100	100	100

New lands (NL_j)	40	100	100	100
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Where project proponents account for cross-commodity production (e.g., where the value for any commodity in the project area ($CP_{j,t}$) in Equation (3) is negative), the project proponent must:

- 1) use the values listed in the “Cross-commodity production default” columns provided in *Table 1. Default values for increased supply (IS_j) and new lands (NL_j)* Table 1 for the commodities displaced from the project area and introduced to the project area to estimate the net change in land area.
- 2) demonstrate that:
 - a) any commodities introduced to the project area are produced within the country.
 - b) production of the commodity is increasing (e.g., use the two most recent reporting time steps in the FAOSTAT dataset² to show that national production of commodity j has increased).
- 3) apply the default values across the entire project area, for all commodities, and across all project activity instances.

Where leakage-only accounting is applied, project proponents may propose alternative values for IS_j or NL_j for displaced agricultural products.³ Where alternative values are used that are less conservative than the default values, the project proponent must provide evidence (e.g., peer-reviewed studies, government reports relevant to the agricultural commodity and the project region) justifying the use of a different value. Project proponents may use more conservative values for IS_j or NL_j without providing such evidence.

Project proponents must provide an estimate of the commodity yield production in units per hectare per year. For subsistence commodities, the land yield ($G_{j,t}$) must reference the yield in the project area during the historical reference period. Subsistence commodities must not be introduced as part of cross-commodity production.

Where regional or national data are not available for fuelwood production, $G_{j,t}$ may be based on average aboveground biomass growth rates published by the IPCC that are applicable to the project region.

For agricultural commodities, yield estimates for new land brought into production or lands spared must be based on data from the geographic domain most aligned with the commodity's

² Available at: <https://www.fao.org/faostat/en/#data>

³ See Appendix 1 for background on how default values for IS_j and NL_j were developed.

market, using information no coarser than the national scale. The total area generating leakage emissions ($INL_{j,t}$) is calculated by summing the area of new land brought into production or spared across all commodities produced in the project area. Where leakage mitigation occurs, the corresponding area will have been adjusted as described in Section 5.2.

The amount of land that is brought into production (where $INL_{j,t}$ is positive) or spared (where $INL_{j,t}$ is negative) outside the project area in year t is calculated as follows:

$$INL_{j,t} = \frac{l_{j,t} \times IS_j \times NL_j}{G_{j,t}} \quad (9)$$

Where:

$INL_{j,t}$	= Area of new land brought into production or spared for commodity j in year t (ha)
$l_{j,t}$	= Net change in production subject to leakage for commodity j in year t (units of production)
IS_j	= Proportion of leakage resulting in change in supply outside the project area for commodity j (%)
NL_j	= Proportion of increased supply from new land brought into production for commodity j (%)
$G_{j,t}$	= Yield on new land brought into production or spared for commodity j in year t (units of production/ha)
t	= 1, 2, 3, ..., t years elapsed since the project start date

Equation (10) aggregates the commodity-specific land impacts into a single value of area expected to generate emissions due to leakage in year t (AL_t). Where the area expected to generate emissions due to leakage in year t is less than zero, leakage mitigation or land sparing from cross-commodity production exceeds displaced production and AL_t is set to zero (projects cannot be credited for a net land sparing impact).

$$AL_t = \text{MAX}(\sum_{j=1}^J INL_{j,t}, 0) \quad (10)$$

Where:

AL_t	= Total area generating leakage emissions in year t (ha)
$INL_{j,t}$	= Area of new land brought into production or spared for commodity j in year t (ha)
J	= Total number of commodities produced in historical reference period
t	= 1, 2, 3, ..., t years elapsed since the project start date

5.4 Step 4: Determine Change in Carbon Stocks in New Lands Brought into Production

This module assumes that new land brought into production is forest land unless the project proponent demonstrates one of the following:

- 1) Forest land is extremely rare, constituting less than 10% of land cover in the country in which the project is located.
- 2) Forest land is effectively protected or otherwise highly unlikely to be converted (e.g., forests are located on steep slopes or only at high elevations). To demonstrate that forests are unlikely to be converted, the annual gross deforestation rate of the country in which the project is located must be less than 0.1% over the past ten years. Global Forest Watch⁴ must be used to substantiate the gross deforestation rate.

Where either of the above conditions are met, the project proponent may proceed as follows:

- a) Identify the relevant ecoregion where the project is located.⁵
- b) Assume that this ecoregion land cover type is converted as the portion of new land brought into production (AL_t).
- c) Estimate the change in biomass carbon stock ($\Delta C_{biomass}$) of all significant carbon pools and the change in soil organic carbon stock (ΔSOC) caused by this conversion.

Where conditions (1)–(2) are not met, the project proponent must assume the new land brought into production is forest land. The change in biomass carbon stocks in the area generating leakage emissions assumes the complete loss of above- and belowground biomass, dead wood, and litter.

The change in carbon stocks in the new land that is brought into production is calculated as follows:

$$\Delta CS = \Delta C_{biomass} + \Delta SOC \quad (11)$$

Where:

- ΔCS = Change in carbon stocks on new lands brought into production (t C/ha)
- $\Delta C_{biomass}$ = Average regional change in carbon stocks of forest or other eligible land cover type in the project region (t C/ha)

⁴ Available at: <https://www.globalforestwatch.org/dashboards/global/>

⁵ The ecoregion is independent of the baseline or project land use and represents the most likely land cover type that is displaced. Olson, D. M., E. Dinerstein, E. D. Wikramanayake, et al. 2001. "Terrestrial Ecoregions of the World: A New Map of Life on Earth." *BioScience* 51 (11): 933–38. [https://doi.org/10.1641/0006-3568\(2001\)051\[0933:TEOTWA\]2.0.CO;2](https://doi.org/10.1641/0006-3568(2001)051[0933:TEOTWA]2.0.CO;2)

ΔSOC = Average regional change in soil organic carbon (SOC) stocks in the project region (t C/ha)

The change in SOC stock is calculated as the difference between initial reference stocks and estimated future, steady-state stocks at the end of 20 years.

$$\Delta SOC = SOC_{REF} \times ((1 - (f_{LU} \times f_{MG} \times f_{IN}))) \quad (12)$$

Where:

ΔSOC = Average regional change in SOC stocks in the project region (t C/ha)
 SOC_{REF} = SOC stock corresponding to the reference condition in the climate region and for the soil type applicable to the region receiving the displaced activity (t C/ha)
 f_{LU} = Relative SOC stock change factors over 20 years for land use applicable to displaced production (dimensionless)
 f_{MG} = Relative SOC stock change factors over 20 years for management practices applicable to displaced production (dimensionless)
 f_{IN} = Relative SOC stock change factors over 20 years for inputs applicable to displaced production (dimensionless)

5.5 Step 5: Determine Leakage Emissions

The leakage emissions from new land brought into production are calculated as follows:

$$LK_t = \left(AL_t \times \Delta CS \times \frac{44}{12} \right) + E_{LM,t} \quad (13)$$

Where:

LK_t = Cumulative leakage up to year t (t CO₂e)
 AL_t = Total area generating leakage emissions in year t (ha)
 ΔCS = Change in carbon stocks on new lands brought into production (t C/ha)
 $E_{LM,t}$ = Total leakage mitigation emissions in year t (t CO₂e)
 t = 1, 2, 3, ..., t years elapsed since the project start date
 $44/12$ = Conversion factor from carbon to carbon dioxide

6 DATA AND PARAMETERS

6.1 Data and Parameters Available at Validation

Data/Parameter	$p_{j,h}$
Data unit	Unit of production, varies by commodity
Description	Production in the project area for commodity j in year h of the historical reference period
Equations	(1)
Source of data	Project proponents must use the first available source from the following list (i.e., where (1) is available, it must be used): 1) Grower records (e.g., management logs, receipts or invoices, logs or files containing machine and/or sensor data), or remote sensing methods (e.g., satellite imagery, manned aerial vehicle footage, drone imagery) where the requisite information on production can be reliably determined using remote sensing 2) Regional (sub-national) average values derived from published census data 3) National average values derived from census data 4) For fuelwood production, average aboveground biomass growth rates published by the IPCC in the <i>2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories, Chapter 4, Volume 4, Tables 4.9–4.11</i>) may be used where regional or national averages are not available. Where using regional or national averages, official government sources must be used where available. Otherwise, published studies may be used to justify such values. Regional and national average values must be derived from all datapoints within the historical reference period for which such data are available, or from the closest datapoint to the end of the historical reference period otherwise. Historical production records must: a) reference the relevant commodity, or ownership class where estimates have been disaggregated by those attributes. b) be substantiated with a signed attestation from the farmer or landowner.

Value applied	Dependent on data source
Justification of choice of data or description of measurement methods and procedures applied	Must be recorded for every year in the historical reference period
Purpose of data	Calculation of foregone production
Comments	Units of production for livestock commodities may be expressed in terms of stocks and may include number of animals, animal units, or forage intake units.

Data/Parameter	<i>H</i>
Data unit	Years
Description	Duration of historical reference period
Equations	(1)
Source of data	N/A
Value applied	Three or the number of years for one complete crop rotation where applicable, whichever is greater
Justification of choice of data or description of measurement methods and procedures applied	Selected by the project proponent
Purpose of data	Calculation of foregone production
Comments	None

Data/Parameter	<i>K</i>
Data unit	Years
Description	Duration of leakage mitigation historical reference period
Equations	(4)

Source of data	N/A
Value applied	Three or the number of years for one complete crop rotation where applicable, whichever is greater
Justification of choice of data or description of measurement methods and procedures applied	Selected by the project proponent
Purpose of data	Calculation of foregone production
Comments	None

Data/Parameter	$OP_{j,k}$
Data unit	Unit of production, varies by commodity
Description	Baseline production in the leakage mitigation area for commodity j in year k of the historical reference period
Equations	(4)
Source of data	Grower records, or remotely sensed data (e.g., satellite imagery, manned aerial vehicle footage, drone imagery) where the requisite information on production can be reliably determined using remote sensing
Value applied	Dependent on data source
Justification of choice of data or description of measurement methods and procedures applied	Must be recorded for every year in the historical reference period
Purpose of data	Calculation of leakage mitigation for foregone production
Comments	Units of production for livestock commodities may be expressed in terms of stocks and may include number of animals, animal units, and forage intake units.

Data/Parameter	EF_i
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Data unit	Dimensionless
Description	Emission factor for fossil fuel i
Equations	(8)
Source of data	IPCC default values from Chapter 3, Volume 2 of IPCC (2019) or peer-reviewed published data on country-specific emission factors. For conservativeness, the upper bound of IPCC values must be used.
Value applied	Fuel-specific
Justification of choice of data or description of measurement methods and procedures applied	Fossil fuel consumption monitored by landowners or entities responsible for mitigation activities
Purpose of data	Calculation of emissions from leakage mitigation activities
Comments	None

6.2 Data and Parameters Monitored

Data/Parameter	r_j
Data unit	%
Description	Annual growth rate of yield for commodity j
Equations	(1), (4)
Source of data	Regional (where available) yield data must be obtained or calculated using published studies or official government statistics referencing the smallest territorial administrative division (e.g., district or municipality) that encompasses all areas of land included in the project activity. Yield data must be disaggregated by the ownership classes for which production data are available. Where published studies and official government statistics are not available for crop commodities, data from FAOSTAT may be used. Alternatively, a default value of 2.5% may be used.
Description of measurement methods	Where using data from FAOSTAT (only applicable for crop commodities), the following procedures must be applied:

and procedures to be applied	1) For year t (years elapsed since the project start date), find the yield for commodity j by selecting Group: Production; Domains: Crops and livestock products; Area: Country where project activity is located; Element: Yield; and Item: Commodity j. 2) Calculate annual growth rate of yield for commodity j as: $r_j = \left(\frac{yield_{j,t}}{yield_{j,t-1}} \right) - 1$
Frequency of monitoring/recording	At each monitoring event
QA/QC procedures to be applied	N/A
Purpose of data	Calculation of baseline commodity production in the project area and leakage mitigation area
Calculation method	N/A
Comments	Where default value is not applied, yield data from the year closest to year t must be used unless otherwise justified. Yield data for livestock commodities is expressed in terms of stocking rate and may include number of animals, animal units, and forage intake units per hectare.

Data/Parameter	$OMP_{j,t}$
Data unit	Units of production
Description	Monitored production in the leakage mitigation area for commodity j in year t
Equations	(5)
Source of data	Grower records (e.g., management logs, receipts or invoices, logs or files containing machine and/or sensor data), or data derived from remote sensing methods (e.g., satellite imagery, manned aerial vehicle footage, drone imagery) where the requisite information on production can be reliably determined using remote sensing
Description of measurement methods and procedures to be applied	N/A

Frequency of monitoring/recording	At each monitoring event
QA/QC procedures to be applied	N/A
Purpose of data	Calculation of leakage mitigation
Calculation method	N/A
Comments	None

Data/Parameter	$E_{fert,t}$
Data unit	t CO ₂ e
Description	Leakage mitigation emissions from incremental increase in fertilizer use
Equations	(7)
Source of data	Calculated using Equations (15)–(21) in VM0047, v1.1 or equivalent equations in more recent version
Description of measurement methods and procedures to be applied	See VM0047.
Frequency of monitoring/recording	At monitoring event
QA/QC procedures to be applied	N/A
Purpose of data	Calculation of leakage mitigation emissions
Calculation method	Calculated using Equations (15)–(21) in VM0047, v1.1 or equivalent equations in more recent version
Comments	None

Data/Parameter	$I S_j$
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Data unit	%
Description	Proportion of leakage resulting in change in supply outside the project area for commodity j
Equations	(9)
Source of data	N/A
Description of measurement methods and procedures to be applied	Default value of 75% for agricultural commodities and 100% for fuelwood Where the default value is not applied (for leakage-only accounting), evidence justifying the use of a different value must be provided (e.g., government-approved or peer-reviewed studies relevant to the agricultural commodity and the project region).
Frequency of monitoring/recording	At each monitoring event
QA/QC procedures to be applied	N/A
Purpose of data	Estimation of area of new land brought into production
Calculation method	N/A
Comments	See Appendix 1 for background on default values. Where an alternative value is applied that is more conservative than the default, the project proponent is not required to justify its use.

Data/Parameter	NL_j
Data unit	%
Description	Proportion of increased supply from new land brought into production for commodity j
Equations	(9)
Source of data	Default value of 40% for agricultural commodities or 100% for fuelwood Where the default value is not applied (leakage-only accounting), evidence justifying the use of a different value must be provided (e.g., government-approved or peer-reviewed studies relevant to the commodity produced and region in which the project is located).

Description of measurement methods and procedures to be applied	N/A
Frequency of monitoring/recording	At each monitoring event
QA/QC procedures to be applied	N/A
Purpose of data	Estimation of area of new land brought into production
Calculation method	N/A
Comments	See Appendix 1 for background on default values. Where an alternative value is applied that is more conservative than the default, the project proponent is not required to justify its use.

Data/Parameter	$G_{j,t}$
Data unit	Units of production/ha
Description	Yield on new lands brought into production or spared for commodity j in year t
Equations	(9)
Source of data	Productivity data must be obtained or calculated using official government statistics or published studies where available. Otherwise, for crop commodities, data from FAOSTAT may be used. Where none of these sources of data are available, or for subsistence commodities, yield in the project area during the historical reference period may be used.
Description of measurement methods and procedures to be applied	Data from the year closest to that for which leakage emissions are being calculated must be used. For non-subsistence commodities, the yield on new land brought into production must reference regional yields for the relevant agricultural commodities subject to foregone production. Where regional data are not available, national data must be used. Where regional and national data are not available, yield in the project area during the historical reference period must be used. Where regional and national data are not available for fuelwood production, yields based on the

	average aboveground biomass growth rates published by the IPCC and applicable to the project region may be used. For subsistence commodities, the yield of new land brought into production must reference the yield in the project area during the historical reference period. Where using data from FAOSTAT (applicable for crop commodities only), the following procedure must be applied. For year t (number of years elapsed since the project start date), find the yield for commodity j by selecting Group: Production; Domains: Crops and livestock products; Area: Country where project activity is located; Element: Yield; Item: Commodity j.
Frequency of monitoring/recording	At each monitoring event
QA/QC procedures to be applied	N/A
Purpose of data	Calculation of area of new land brought into production
Calculation method	N/A
Comments	Where yield data are not available for year t, data from the closest available year to year t must be used unless otherwise justified. Yield on new lands brought into production for livestock commodities may be expressed in terms of stocking rate and may include number of animals, animal units, and forage intake units.

Data/Parameter	$\Delta C_{biomass}$
Data unit	t C/ha
Description	Average regional change in carbon stocks of forest or other eligible land cover type
Equations	(11)
Source of data	Published studies with data from the country in which the project is located where such data are available. Otherwise, derived from Table 3A.1.4 of the <i>IPCC Good Practice Guidance for Land Use, Land-Use Change and Forestry</i> (IPCC 2003).
Description of measurement methods	Where forest land is assumed to be displaced (see Section 5.4):

and procedures to be applied

$\Delta C_{biomass}$ must reference the following data sources as available in the order listed (e.g., where two data sources are available, the data source nearest the top of the list must be used):

- 1) Mean carbon stocks in forest biomass in the region
- 2) Mean carbon stocks in forest biomass in the country where the project is located, obtained from published studies or official government statistics
- 3) Mean carbon stocks in forest biomass derived from aboveground biomass estimates in Table 3A.1.4 of the *IPCC Good Practice Guidance for Land Use, Land-Use Change and Forestry* (2003)

Determined based on data available at the time of the assessment and reflecting area-weighted mean stocks in above- and belowground biomass, dead wood, and litter pools at the nearest date to the assessment date.

Where biomass is presented in tonnes of dry biomass per hectare, carbon stocks must be determined using a carbon fraction default value of 0.47.

Where only aboveground biomass stocks are reported, belowground biomass stocks should be estimated using allometric equations from Table 4.A.4 of IPCC (2003), unless verifiable information is provided to justify a different value.

Where data include tree biomass only, the carbon stocks in dead wood and litter must be estimated by multiplying the sum of aboveground and belowground biomass stocks by a factor of 1.1.⁶

Where a non-forest land type is assumed to be displaced (see Section 5.4):

$\Delta C_{biomass}$ for the identified ecosystem must reference the following data sources as available in the order listed (e.g., where two data sources are available, the data source nearest the top of the list must be used):

- a) Region-specific biomass or carbon stock data for the ecosystem
- b) National-level studies or official statistics for the ecosystem type
- c) IPCC default factors or tables relevant to the ecosystem

Ensure that above- and belowground biomass, dead wood, litter, soil organic carbon, and all other significant pools are included.

⁶ Source: CDM AR-TOOL15: *Estimation of the Increase in GHG Emissions Attributable to Displacement of Pre-project Agricultural Activities in A/R CDM Project Activity*, v2.0

	Convert all biomass to carbon using a default carbon fraction of 0.47, unless more specific ecosystem data are available and their use is justified.
Frequency of monitoring/recording	At the time of each monitoring event
QA/QC procedures to be applied	N/A
Purpose of data	Calculation of leakage emissions
Calculation method	N/A
Comments	None

Data/Parameter	SOC_{REF}
Data unit	t C/ha
Description	Soil organic carbon (SOC) stock corresponding to the reference condition in the climate region and for the soil type applicable to the region receiving the displaced activity
Equations	(12)
Source of data	Table 2.3 in Chapter 2, Volume 4 of the 2019 <i>Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories</i> or most recent version
Description of measurement methods and procedures to be applied	Where soils vary across the region of production, a weighted average (based on relative commodity production) or the highest, most conservative value may be used.
Frequency of monitoring/recording	At each monitoring event
QA/QC procedures to be applied	N/A
Purpose of data	Calculation of change in carbon stocks on new lands brought into production
Calculation method	N/A

Comments	None
Data/Parameter	f_{LU} f_{MG} f_{IN}
Data unit	Dimensionless
Description	Relative SOC stock change factors over 20 years for land use applicable to displaced production Relative SOC stock change factors over 20 years for management practices applicable to displaced production Relative SOC stock change factors over 20 years for inputs applicable to displaced production
Equations	(12)
Source of data	- For displaced crops, referencing tree crops in the case of fuelwood: Table 5.5 in Chapter 5 in Volume 4 of the <i>2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories</i> - For displaced grazing: Table 6.2 in Chapter 6 in Volume 4 of the <i>2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories</i>
Description of measurement methods and procedures to be applied	N/A
Frequency of monitoring/recording	At each monitoring event
QA/QC procedures to be applied	N/A
Purpose of data	Calculation of leakage emissions
Calculation method	N/A
Comments	None

7 REFERENCES

Alexandratos, N., and J. Bruinsma. 2012. *World Agriculture towards 2030/2050: The 2012 Revision*. Food and Agricultural Organization Agricultural Development Economics Division. <http://www.fao.org/docrep/016/ap106e/ap106e.pdf>

Fuglie, K. O., and A. Nin-Pratt. 2012. "Agricultural Productivity: A Changing Global Harvest." In *2012 Global Food Policy Report*, edited by International Food Policy Research Institute. <http://dx.doi.org/10.2499/9780896295537>

IPCC. 2003. *Good Practice Guidance for Land Use, Land-use Change and Forestry*. Institute for Global Environmental Strategies (IGES). <https://www.ipcc.ch/publication/good-practice-guidance-for-land-use-land-use-change-and-forestry/>

IPCC. 2019. *2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories Volume 4*. IPCC. <https://www.ipcc-nggip.iges.or.jp/public/2019rf/index.html>

Roberts, M. J., and W. Schlenker. 2013. "Identifying Supply and Demand Elasticities of Agricultural Commodities: Implications for the US Ethanol Mandate." *American Economic Review* 103 (6): 2265–95. <https://www.doi.org/10.1257/aer.103.6.2265>

Verra. 2014. *AFOLU Project Market Leakage: Production Approach. Public consultation document 29 May 2014*. <https://verra.org/wp-content/uploads/imported/methodologies/AFOLU-Project-Market-Leakage-Public-Consultation.pdf>

APPENDIX 1: BACKGROUND ON DEFAULT VALUES

Growth Rate of Production

The default value for growth rates of production was developed based on reports from the International Food Policy Research Institute (IFPRI) and Food and Agriculture Organization (FAO). This default value based on the growth rates of production is applied within this module to serve as a proxy for growth rates in yields. Analysis from IFPRI indicates that global average agricultural growth rates over the past 40 years have remained less than 2.5% per decade, with values ranging from 2.08% to 2.42% (Fuglie and Nin-Pratt 2012). Reports from FAO confirm that this is a conservative default value for growth rates. FAO predicts that agricultural growth rates in developing countries will decrease in the coming decades with an average value of 1.6% from 2007 to 2030 and 0.9% from 2030 to 2050 (Alexandratos and Bruinsma 2012). Therefore, 2.5% was selected as a conservative default value for commodity production. Project proponents may justify using regional or national values instead where such data are relevant and available.

Proportion of Leakage Resulting in Increased Supply

The default value for the proportion of leakage resulting in increased supply of agricultural commodities outside the project area was developed considering elasticities of supply and demand as well as observed prices and quantities to develop constant elasticity of substitution (CES) demand, supply, and reduced supply curves. The analysis for developing the conservative default value used elasticities of supply and demand across commodities and countries from peer-reviewed economic studies. Data on exogenous prices and quantities of commodities in specific geographic regions were gathered from several sources including the USDA Economic Research Service (ERS), Food and Agricultural Organization of the United Nations (FAO), Federal Reserve Economic Data (FRED), and Trading Economics. Elasticity data on agricultural commodities and forest products indicate that no commodities experienced perfectly inelastic supply or demand. Averaging across global commodities, production losses occurring within the project would result in a 13–72% increase in supply elsewhere. Therefore, 75% was selected as a conservative default value.

Proportion of Increased Supply Coming from New Land Brought into Production

The default value for the increased supply coming from new land brought into production was developed based on data and reports from the FAO. Such studies indicate that globally only 10% of increases in agricultural production will come from new land being brought into production (Alexandratos and Bruinsma 2012). Some sub-regions such as Latin America and the Caribbean may experience 40% of future supply coming from new land being brought into production. While these studies provide predictions regarding long-term trends in production, other studies on short-term trends result in other conclusions. Roberts and Schlenker (2013) provide evidence that shorter-term price

shocks do not follow these longer-term trends and the increase in production is primarily made up by bringing new land into production.

The default value has been developed based on the assumption that projects will have longer-term impacts on commodity markets. Based on FAO data, 40% was selected as a conservative threshold for the default value for agricultural commodities. This default value may be updated in the future where projects are shown to have shorter-term impacts on commodity markets.

Conservative Assumptions Used

Use of the above default values is further supported by the following conservative assumptions that have been employed elsewhere in this module:

- 1) New lands brought into production are assumed to be forested lands.
- 2) Positive leakage effects from ARR activities that provide timber to the market and reduce harvesting on existing forested lands are not included. Similarly, increases in production in leakage mitigation areas that occur after leakage emissions are calculated are not included.
- 3) Leakage calculations conservatively ignore that new lands brought into production outside the host country would reduce the amount of new land brought into production within the host country and would reduce leakage emissions.⁷
- 4) Leakage estimated in this module does not decrease with time. However, actual leakage effects may be expected to decrease as the proportion of carbon projects and the impact on market supply and prices from foregone production increase over time.

⁷ Leakage is limited to domestic emissions in the *VCS Standard*.

DOCUMENT HISTORY

Version	Date	Changes
v1.0	28 Sep 2023	Initial version released
v1.1	13 Jan 2026	• Module renamed to <i>VMD0054 Estimating Leakage from the Displacement of Agricultural Activities</i> • Updates to enable use with <i>VM0032</i> and <i>VM0042</i> • Updates to leakage mitigation area requirement • Addition of cross-commodity production accounting • Inclusion of other land cover types in the regional average of biomass • Removal of original Appendix 1 (Leakage Example), to be provided as separate guidance • General copyediting